

LABORATORY RECORD

EXP NO: 3 IMPLEMENTING ASSET MANAGEMENT WITH HYPERLEDGER FABRIC

AIM

To implement an Asset Management application using Hyperledger Fabric by creating, reading, updating, transferring, and deleting assets through JavaScript chaincode and verifying the transactions on the blockchain ledger.

PROCEDURE

Step 1: Navigate to the Fabric Test Network cd

```
/mnt/c/Users/Sivarangini/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Deploy the Asset Transfer Chaincode

```
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
```

Step 4: Initialize the Ledger

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls -cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "InitLedger", "Args": []}'
```

Step 5: Query All Assets peer chaincode query -C mychannel -n basic -c

```
'{"Args": ["GetAllAssets"]}'
```

Step 6: Create a New Asset

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls -cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "CreateAsset", "Args": ["asset55", "Silver", "22", "Sivarangini", "1800"]}'
```

Step 7: Read the Asset Details peer chaincode query -C mychannel -n basic -c

```
'{"Args": ["ReadAsset", "asset55"]}'
```

Step 8: Update Asset Information

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls -cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "UpdateAsset", "Args": ["asset55", "Titanium", "24", "Sivarangini", "2300"]}'
```

Step 9: Transfer Asset Ownership

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls -cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "TransferAsset", "Args": ["asset55", "Karthik"]}'
```

Step 10: Verify the Updated Asset

```
peer chaincode query -C mychannel -n basic -c '{"Args": ["ReadAsset", "asset55"]}'
```

Step 11: Delete the Asset

```
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls -
cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "DeleteAsset", "Args": ["asset55"]}'
```

Step 12: Verify Asset Deletion

```
peer chaincode query -C mychannel -n basic -c '{"Args": ["ReadAsset", "asset55"]}'
```

Step 13: Stop the Network

```
./network.sh down
```

OUTPUT

```
sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ ./network.sh up createChannel -ca Creating channel
'mychannel'...
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Network Started Successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ ./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-
transferbasic/chaincode-javascript Packaging Chaincode...
Installing Chaincode on peer0.org1 and peer0.org2...
Approving Chaincode definition for Org1 and Org2...
Committing Chaincode definition to channel 'mychannel'...
Chaincode committed successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel
-n basic -c '{"function": "InitLedger", "Args": []}'
2026-08-20 11:20:12.802 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001
Chaincode invoke successful. result: status:200
Ledger Initialized Successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode query -C mychannel -n basic -c '{"Args":
["GetAllAssets"]}' [
  {"ID":"asset1", "Color":"blue", "Owner":"Tomoko", "Size":5, "AppraisedValue":300},
  {"ID":"asset2", "Color":"red", "Owner":"Brad", "Size":5, "AppraisedValue":400}
] Assets Retrieved
Successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C
mychannel -n basic -c '{"function":"CreateAsset", "Args":["asset55", "Silver", "22",
"Sivarangini",
"1800"]}' Asset asset55 created
successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode query -C mychannel -n basic -c '{"Args":
["ReadAsset", "asset55"]}'
{"ID": "asset55", "Color":"Silver", "Size":22, "Owner": "Sivarangini",
```

```

"AppraisedValue":1800} sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-
dev/fabric-samples/testnetwork$ peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C
mychannel -n basic -c '{"function": "UpdateAsset", "Args":["asset55", "Titanium",
"24", "Sivarangini", "2300"]}' Asset Updated Successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C
mychannel -n basic -c '{"function": "TransferAsset", "Args":["asset55", "Karthik"]}'
Ownership transferred successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode query -C mychannel -n basic -c
'{"Args":["ReadAsset", "asset55"]}'
{"ID":"asset55", "Color": "Titanium", "Size": 24, "Owner": "Karthik",
"AppraisedValue":2300}

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C
mychannel -n basic -c '{"function":"DeleteAsset", "Args":["asset55"]}' Asset Deleted
Successfully

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ peer chaincode query -C mychannel -n basic -c
'{"Args":["ReadAsset", "asset55"]}' Error: asset55 does not exist

sivarangini@LAPTOP-KV7M51GT:/mnt/c/Users/Sivarangini/fabric-dev/fabric-
samples/testnetwork$ ./network.sh down Stopping network...
Fabric Network Stopped Successfully

```

RESULT

The Asset Management application was successfully implemented using Hyperledger Fabric. The asset lifecycle operations, including asset creation, querying, updating, ownership transfer, and deletion, were executed successfully. The transactions were recorded on the blockchain ledger, demonstrating secure and immutable asset management in a permissioned blockchain network.